PROJECT RESURRECTION - IEEE SBUB

Project Documentation Team 1

IEEE SBUB · Open Day · First-Year Projects Exhibition

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0. Project Identity

Project title: Air Theremin

Team name: sʌnth

Team members (full name, group): Sidra Chalouche G01, Tesnim Azzeb G01, Anais Yakoubi G01, Amina Nefida G03, Imene El Batoul Abdaoui G04, Wissem Zaimeddine G09

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Category (Electronics / Arduino / Programming / Robotics / Other): Electronic, heavily reliant on Arduino but not purely (includes other components)

Date of submission: Monday, November 2nd 2026

1. Project Summary

Air Theremin is an Electronic Musical Instrument that uses ultrasonic sensors to detect hand movements and convert them into sounds, allowing the user to control both pitch and volume without any physical contact. It is designed for Tech Enthusiasts interested in Interactive Music.

2. The Idea Behind It (Concept)

Problem or inspiration: what problem does it solve, or what inspired you?

Our inspiration came from the combination of Artistic Expression and Technological Innovation, as well as fascination with how something as simple as hand movements can control sound.

Objectives: what did you want to achieve?

We wanted to achieve a Fun and Interactive Musical Instrument that is pleasant to listen to and able to produce Musical Notes and even recognized melodies with certain hand movements.

Why this approach? Why this solution over others?

It encourages L1 students to experience bringing Old Components together to Create something New and Enjoy it.

Resurrection angle: what did you rebuild, repair, reuse or bring back to life?

Reigniting the concept of the Russian Physicist Leon Theremin, which dates back to 1928, by turning it into a Modern Electrical Musical Instrument.

3. How It Works (Theory & Operation)

Working principle, in your own words:

An Air Theremin uses two ultrasonic sensors to detect hand movements without physical contact. One sensor controls the pitch by measuring the distance of the hand, while the other controls the volume. An Arduino processes these readings and generates sound accordingly, allowing the user to play music through hand gestures.

Key concepts or laws used (e.g. Ohm's law, PWM, logic gates, sensors):

Frequency and period:

$$f = \frac{1}{T} \qquad \Longleftrightarrow \qquad T = \frac{1}{f}$$

Half period (used for mapping the note):

$$t_{\text{half}} = \frac{T}{2} = \frac{1}{2f}$$

Echo ranging (measuring hand distance, HC-SR04):

$$d = \frac{v \cdot t}{2}, \qquad v \approx 343\ \mathrm{m/s}$$

Ohm's law (sizing resistors):

$$V = R \cdot I \qquad \Longleftrightarrow \qquad R = \frac{V}{I}$$

Block diagram or flowchart:

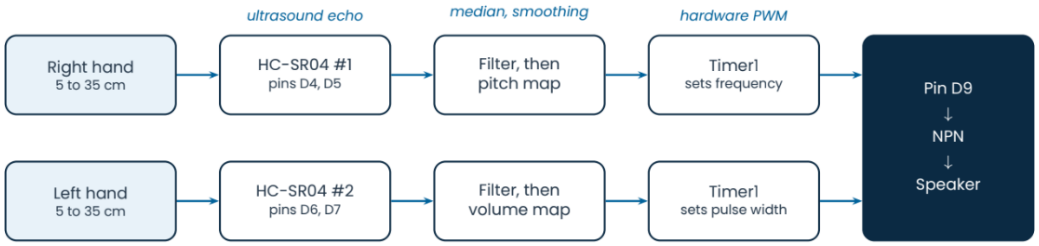


Figure 1: Block Diagram of the Air Theremin System (done by our Mentor)

4. Components & Tools

Component / Software	Quantity	Role in the project
Arduino Uno + USB cable	1	Processes sensor data and generates sound
HC-SR04 ultrasonic sensor	2	One measures the hand distance for pitch, the other for volume
1 kΩ resistor	1	Limits base current
100 Ω resistor	1	Limits speaker current
Speaker	1	Produces sound
Motor driver	1	Drives the speaker with enough current
Breadboard + jumper wires	1 + several	Connects the components
Arduino IDE / Tinkercad	–	Code and simulation

5. Design & Implementation

Circuit schematic / mechanical design / software architecture:

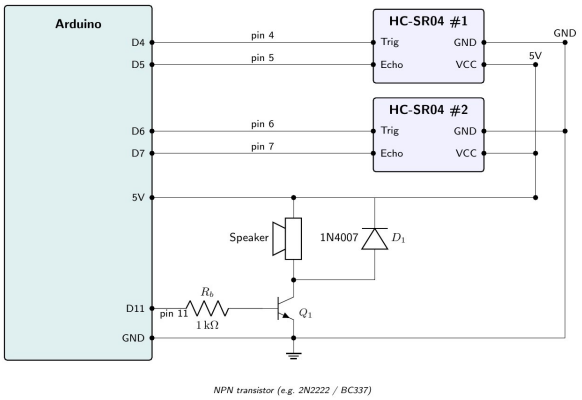


Figure 2: Circuit schematic of the Air Theremin (done by our Mentor)

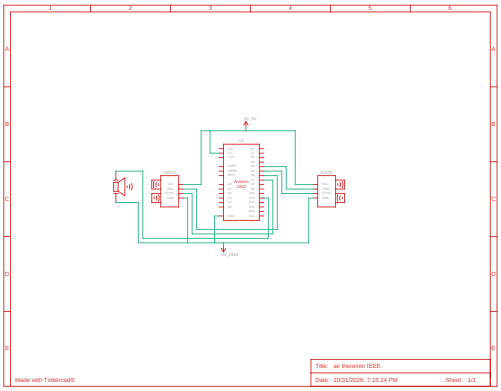


Figure 3: PCB layout of the Air Theremin (done by Tesnim)

Main steps of the build:

1. Set up the Arduino Uno and Arduino IDE.
2. Connect an LED and program it to blink.
3. Read digital inputs using a push button.
4. Connect an LDR and experiment with analog readings and PWM.
5. Connect the two HC-SR04 ultrasonic sensors and measure distances.
6. Generate sound using Arduino Timer1.
7. Assemble the complete circuit, connect the speaker and test the system.

Code or program:

```
#include <Arduino.h>
const byte TRIG_P = 4, ECHO_P = 5;   // pitch
const byte TRIG_V = 6, ECHO_V = 7;  // volume
const byte BUZ = 9;
const float MIN_CM = 5, MAX_CM = 50;
const float F_LOW = 200, F_HIGH = 2000;
const float NO_HAND_CM = 60;
const unsigned long TIMEOUT_US = 40000;
float fSmooth = F_LOW, vSmooth = 0;

float readCm(byte trig, byte echo) {
  digitalWrite(trig, LOW);  delayMicroseconds(2);
  digitalWrite(trig, HIGH); delayMicroseconds(10);
  digitalWrite(trig, LOW);
  unsigned long d = pulseIn(echo, HIGH, TIMEOUT_US);
  if (d == 0) return -1;
  return d * 0.0343 / 2;
}


void sound(float freq, float vol) {        // vol from 0 to 1
  uint16_t top = (uint16_t)(2000000.0 / freq) - 1;
  ICR1  = top;                             // sets the frequency
  OCR1A = (uint16_t)(top * 0.5 * vol);     // sets the duty cycle (volume)
}

void setup() {
  Serial.begin(9600);
  pinMode(TRIG_P, OUTPUT); pinMode(ECHO_P, INPUT);
  pinMode(TRIG_V, OUTPUT); pinMode(ECHO_V, INPUT);
  pinMode(BUZ, OUTPUT);

  // --- Timer1 setup: make D9 output a PWM wave with adjustable frequency ---
  TCCR1A = 0;                     // start from a clean slate
```

```
  TCCR1B = 0;

  TCCR1A |= _BV(WGM11);          // mode 14 (part 1): fast PWM, top = ICR1
  TCCR1B |= _BV(WGM12) | _BV(WGM13);   // mode 14 (part 2)
  TCCR1A |= _BV(COM1A1);         // connect the PWM to pin D9
  TCCR1B |= _BV(CS11);           // tick speed: 16 MHz / 8 = 2 MHz

  ICR1  = 1999;                  // top value: sets the frequency (1 kHz at start)
  OCR1A = 0;                     // switch-off point: sets the volume (silent at start)
}

void loop() {
  float dp = readCm(TRIG_P, ECHO_P);
  delay(20);                                  // let the echo die out
  float dv = readCm(TRIG_V, ECHO_V);
  delay(20);

  Serial.print(dp); Serial.print("␣␣"); Serial.println(dv);

  // No hand on either sensor = silence
  if (dp < 0 || dv < 0 || dp > NO_HAND_CM || dv > NO_HAND_CM) {
    OCR1A = 0;
    return;
  }

  dp = constrain(dp, MIN_CM, MAX_CM);
  dv = constrain(dv, MIN_CM, MAX_CM);

  // Closer to the pitch sensor = higher note
  float freq = F_HIGH - (dp - MIN_CM) / (MAX_CM - MIN_CM) * (F_HIGH - F_LOW);
  // Closer to the volume sensor = louder
  float vol  = 1.0 - (dv - MIN_CM) / (MAX_CM - MIN_CM);

  // Smoothing
  fSmooth = 0.7 * fSmooth + 0.3 * freq;
  vSmooth = 0.7 * vSmooth + 0.3 * vol;

  sound(fSmooth, vSmooth);
}
```

Photos of the real prototype:

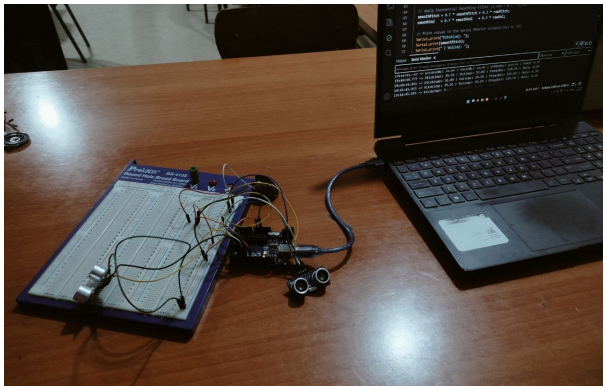


Figure 4: 1st Prototype

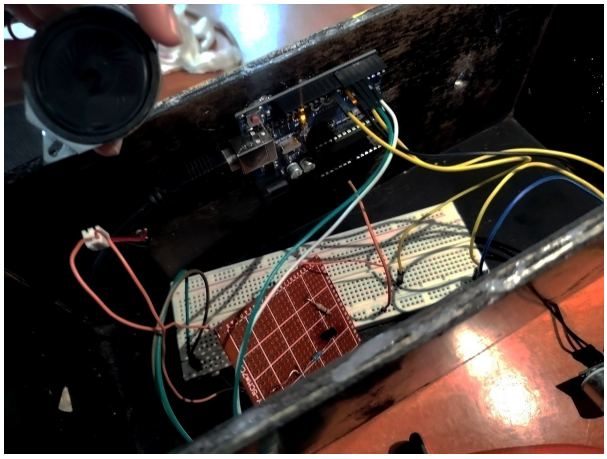


Figure 5: 2nd Prototype

6. Completion Status

Feature / Part	Status (Done / Partial / Not done)	Notes
Arduino setup and programming	Done	Arduino Uno programmed using Arduino IDE
Ultrasonic sensors	Done	Two sensors used for hand detection
Pitch control	Done	Right hand controls frequency
Volume control	Done	Left hand controls loudness
Speaker output	Done	Produced sound
Complete assembly	Done	All components integrated
Painting	Done	Case painted
Soldering	Done	Components soldered on the board
Testing	Done	All features tested and working

Overall completion: 100 %

What is missing, and why?

Nothing is missing, all planned features work.

7. Complexity & Challenges

What was the hardest part?

The most challenging part was figuring out how to program each component and make sure they all worked together properly.

Problems you faced and how you solved them:

We tested 4 ultrasonic sensors but none of them worked. Then we found out that the code had a small timeout of 3000 μs that was cutting the signal. We fixed it by removing the timeout. Another problem we faced was the interference between the two ultrasonic sensors when they operate at the same time, which can affect the accuracy of their readings, hence affecting the quality of the produced sound. We reduced that by positioning them at least 10-15 cm apart and angling them away from each other.

What did you design or code yourselves, and what did you take from existing sources?

We relied on existing Arduino libraries and took the core code from a previous module, modifying and adapting it to meet the specific needs of our project.

8. Testing & Results

How did you test it?

We tested the project by moving our hands at different distances from the two ultrasonic sensors and observing how the sound changed. We checked whether the right-hand sensor accurately controlled the pitch and the left-hand sensor controlled the volume, adjusting the code when necessary. We also added a feature that maps specific hand distances to musical notes, with each semitone corresponding to approximately 1.25 cm. For example, placing the hand at 5 cm produces A5 (880 Hz), at 20 cm produces A4 (440 Hz), and at 35 cm produces A3 (220 Hz). We tested it by attempting to play ”Twinkle, Twinkle, Little Star” and checking whether the system could reproduce the correct sequence of notes and produce a recognizable melody.

Results obtained (measurements, screenshots, short video link):

A short demonstration video of the working prototype is available here:

► Watch the Air Theremin Demo

Click the button to open the video on Google Drive

Limitations:

- The ultrasonic sensor's detection range is limited to 50 cm.
- When both ultrasonic sensors operate simultaneously, their signals may interfere with each other, causing inaccurate distance readings and reduced sound quality.
- The speaker may not deliver clear or high quality sound.

9. Team Contribution

Member	Tasks done
Tesnim Azzeb	Circuit wiring • Soldering • Code modification • Troubleshooting and debugging • Functional testing • Final assembly
Sydra Chalouche	Circuit wiring • Soldering • Final assembly • Documentation scripting
Wissem Zaimeddine	Circuit wiring • Soldering • Final assembly • Report typesetting in LaTeX
Imene El Batoul Abdaoui	Circuit wiring • Soldering • Painting • Final assembly
Anaïs Yakoubi	Circuit wiring • Soldering • Final assembly • Functional testing
Amina Nefida	Circuit wiring • Soldering • Painting • Final assembly

Acknowledgements

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- **Youcef Harmouni** (our Mentor): for his mentoring and guidance throughout the process, for providing the building guide (see References), and for the PCB layout.
- **Amine Khammar**: for the exterior design and painting of the project.

Thank you for your time, your support and your valuable work.

10. Future Improvements

What would you add or improve with more time or budget?

- Add an LED bar to visually display the notes being played.
- Introduce different musical scales, such as major and pentatonic.
- Add a vibrato effect to make the sound more expressive.
- Improve sensor accuracy and stability to make hand movements more precise.
- Improve the speaker's sound quality.

11. References

Books, websites, tutorials, datasets, libraries and AI tools used (be transparent):

- Air Theremin Build Guide by Youcef Harmouni, IEEE Student Branch, Boumerdes University.
- Arduino Uno documentation.
- Arduino IDE documentation.
- Arduino Language Reference.
- https://www.tinkercad.com for simulation.
- Claude by Anthropic for LaTeX codes.
- https://www.overleaf.com for documentation.

Declaration: We confirm that this work is ours and that all sources are cited.

Signature of team leader: 